



CSEE4119: Computer Networks

Introduction & Basics

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Slides credited to J.F Kurose and K.W. Ross and their textbook "Computer Networking: A Top-Down Approach"

Overview/Roadmap

- What *is* the Internet? What *is* a protocol?
- **Network edge**: hosts, access network, physical media
- **Network core**: packet/circuit switching, internet structure
- **Performance**: loss, delay, throughput
- Protocol layers, service models

The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

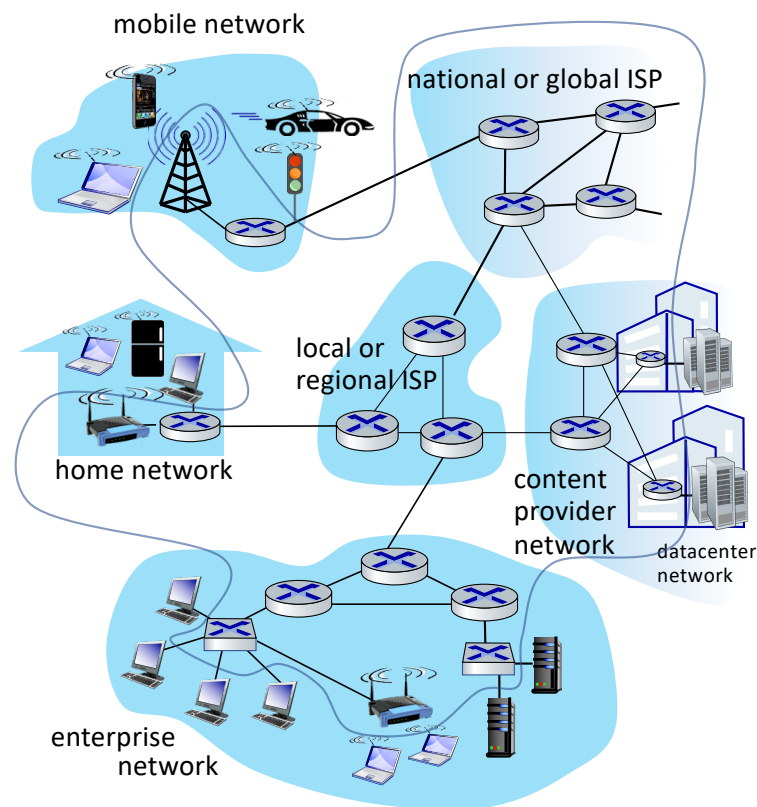
Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*



Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use



Security Camera



Slingbox: remote
control cable TV



Web-enabled toaster +
weather forecaster



cars



bikes



scooters



Internet phones



Gaming devices



sensorized,
bed
mattress



AR devices

Fitbit

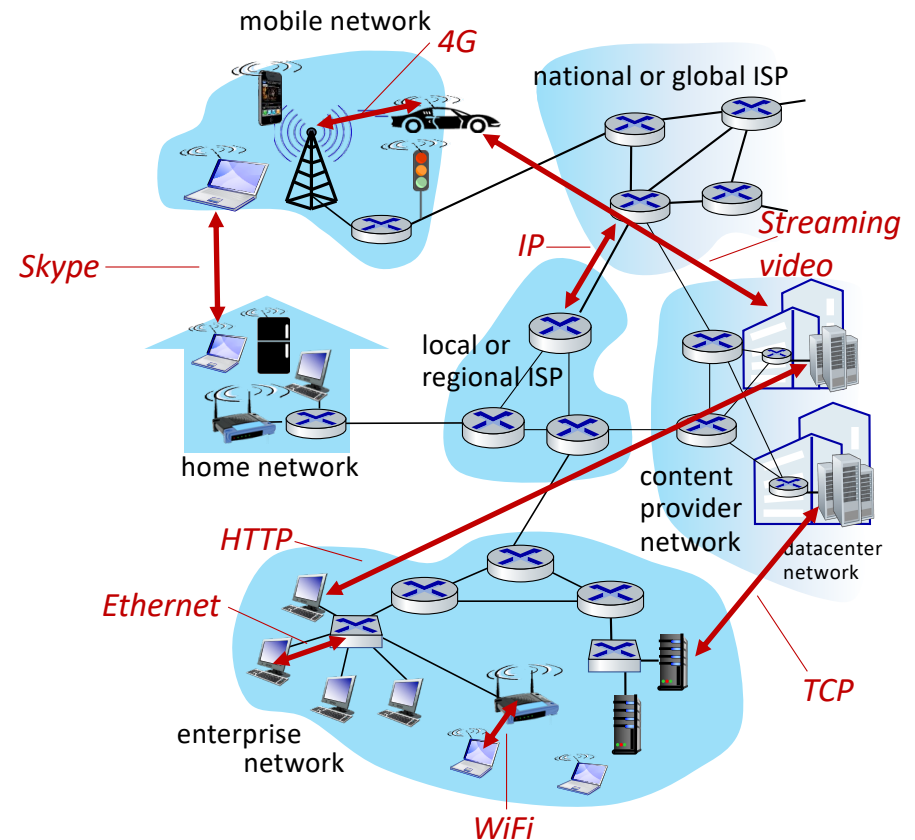


diapers

Others?

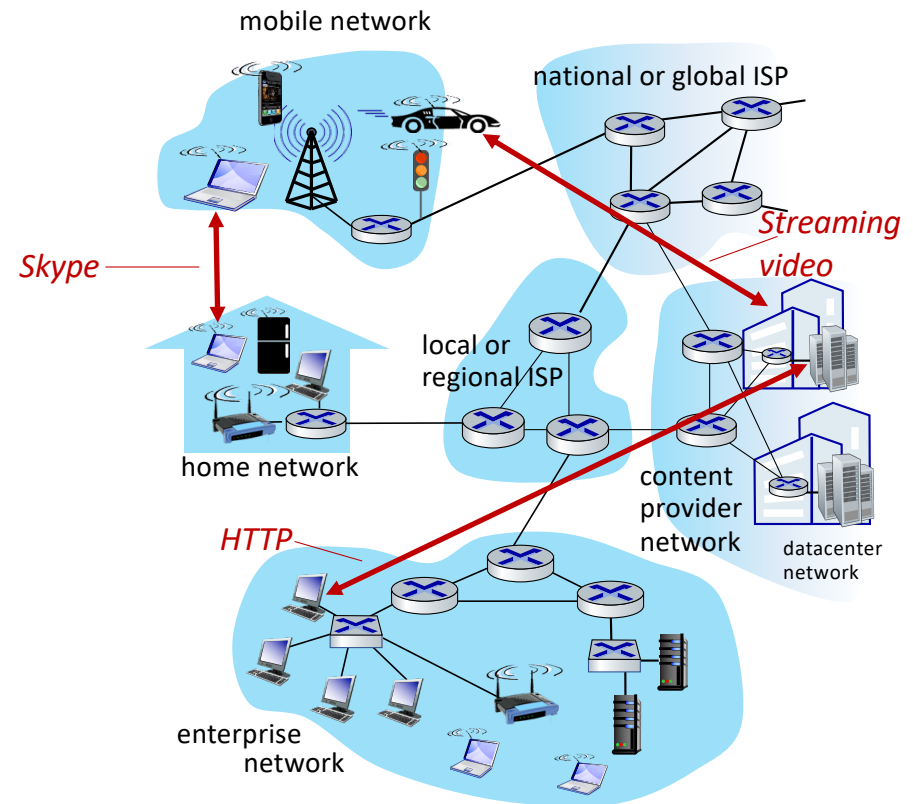
The Internet: a “nuts and bolts” view

- *Internet: “network of networks”*
 - Interconnected ISPs
- *protocols are everywhere*
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

... specific messages sent

... specific actions taken
when message received,
or other events

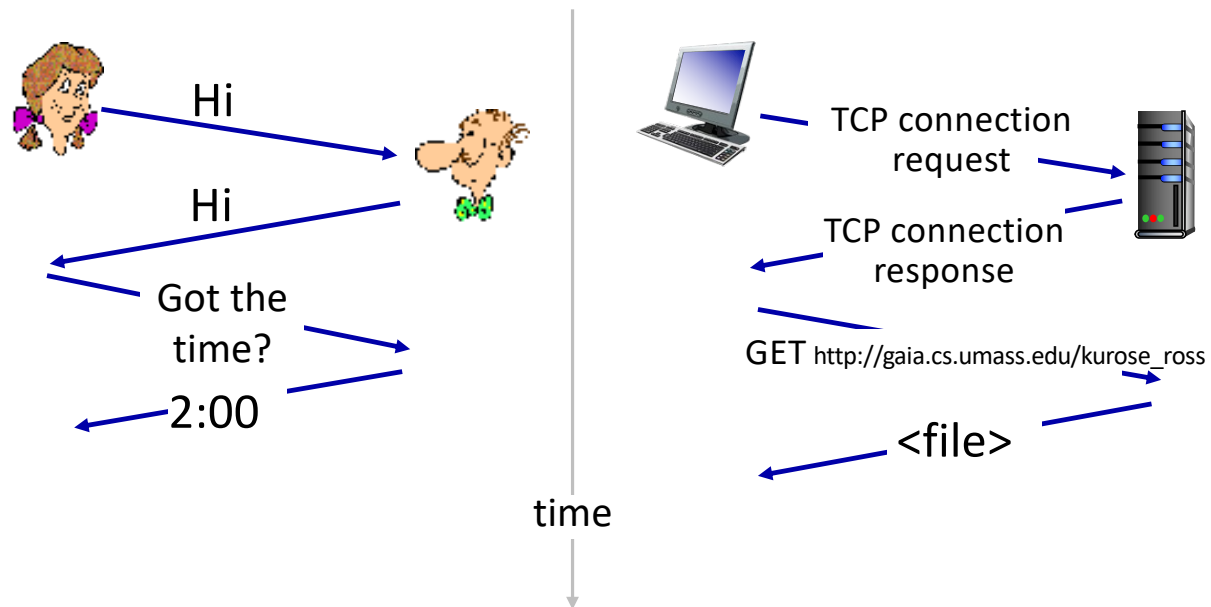
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

*Protocols define the **format, order of messages sent and received** among network entities, and **actions taken** on message transmission, receipt*

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

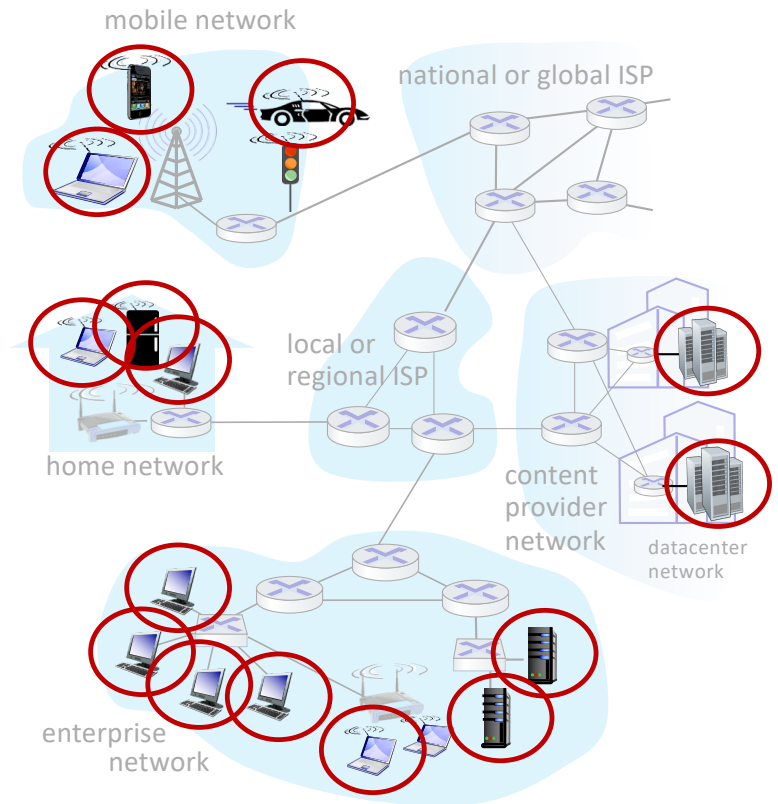
Overview/Roadmap

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A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



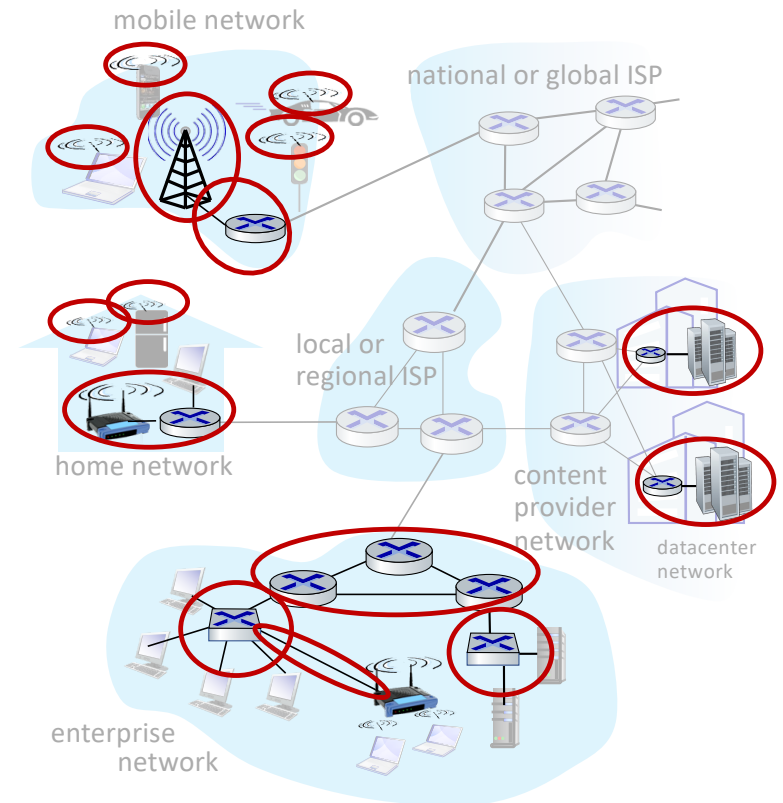
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Access networks, physical media:

- wired, wireless communication links



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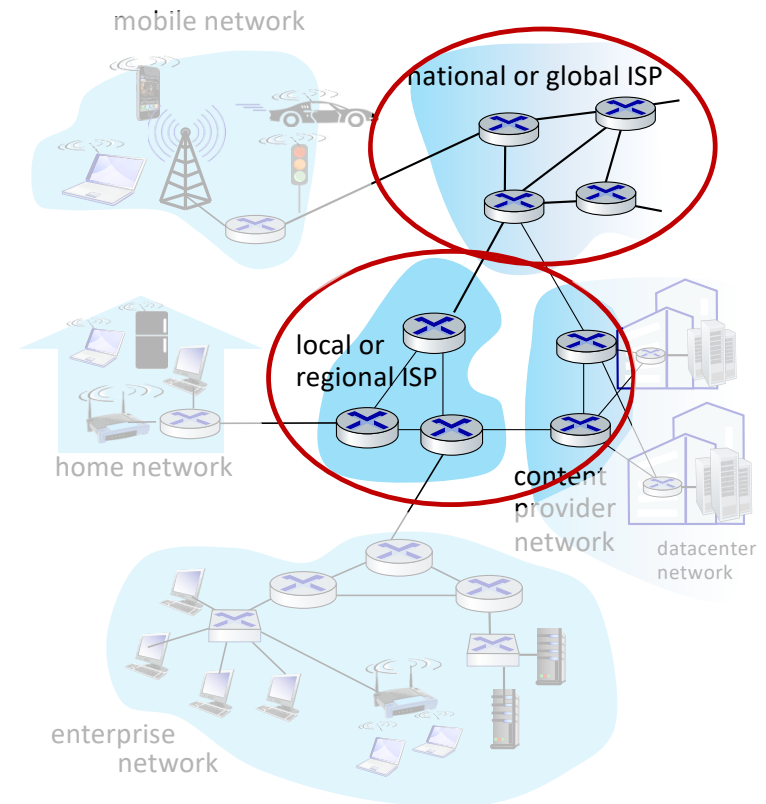
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Network core:

- interconnected routers
- network of networks



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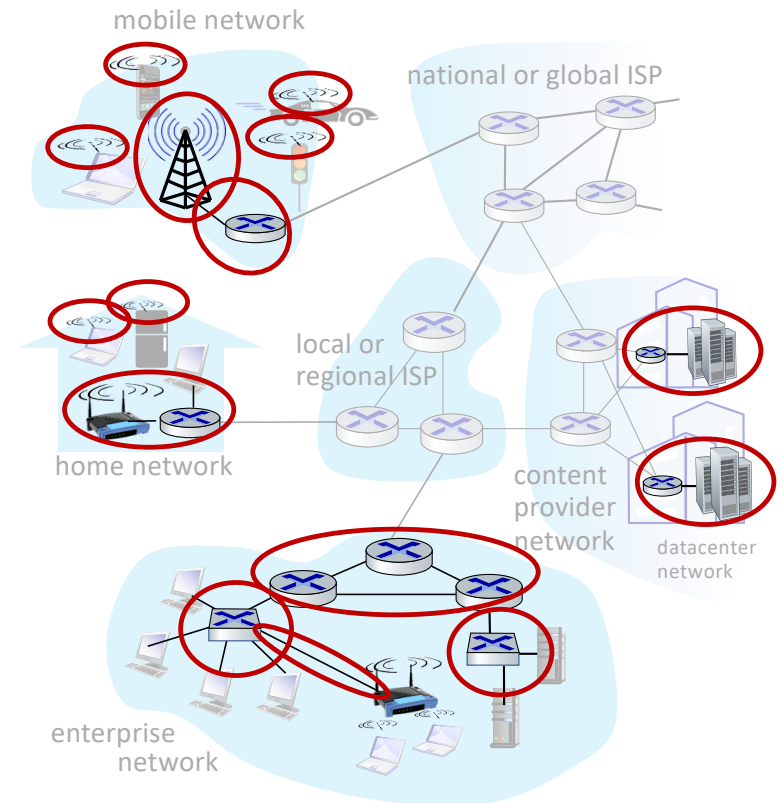
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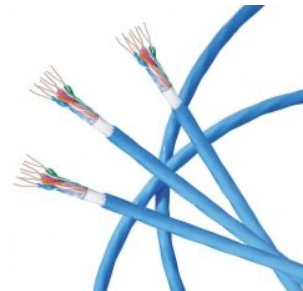


Links: physical media

- **physical link:** what lies between transmitter & receiver
 - **Wired link (guided media)**
 - signals propagate in solid media: copper, fiber, coax
 - **Wireless link (unguided media)**
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet
- Cheaper



Links: physical media

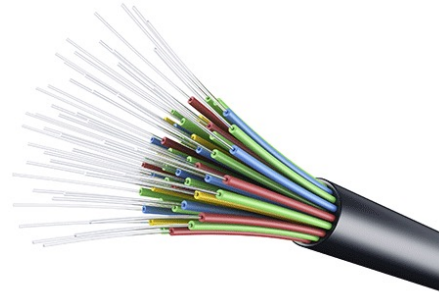
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel
- Better shielding from crosstalk, support longer length



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: physical media

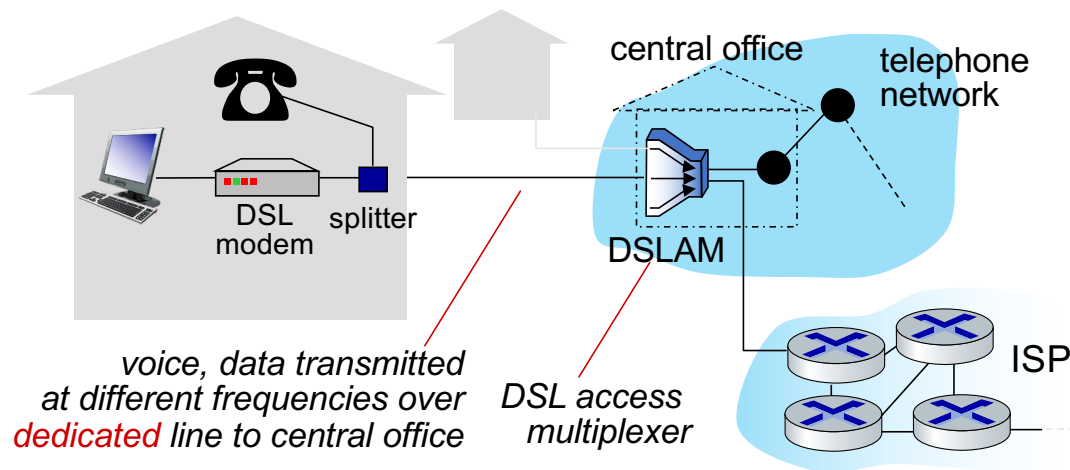
Wireless radio

- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- Broadcast or directional, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

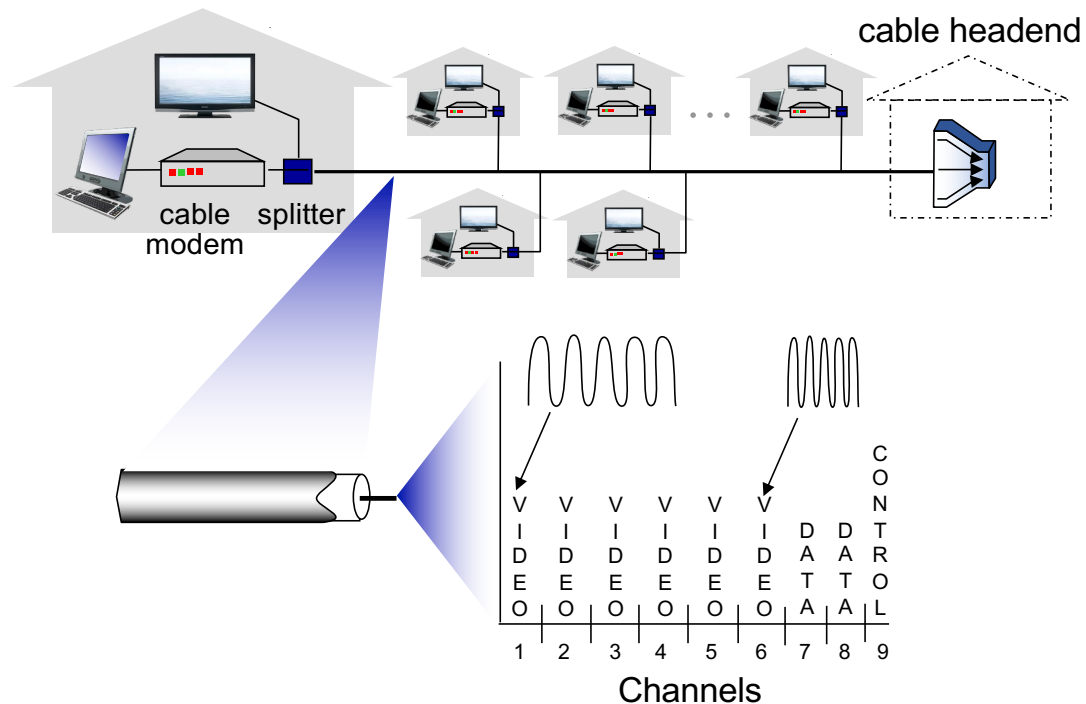
- wireless LAN (Wi-Fi)
 - 10-100's Mbps; 10's of meters
- wide-area (e.g., 4G/5G cellular)
 - 10's Mbps (4G) over ~10 Km
- Bluetooth: cable replacement
 - short distances, limited rates
- terrestrial microwave
 - point-to-point; 45 Mbps channels
- satellite
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

Wired access network #1: digital subscriber line (DSL)



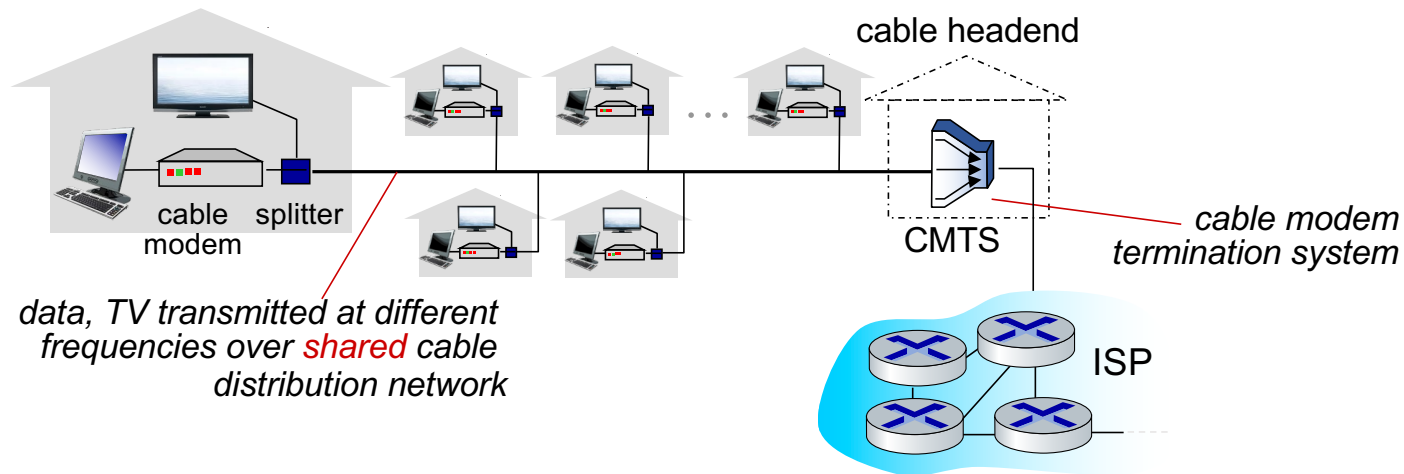
- use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Wired access network #2: cable-based access



frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Wired access #2: cable-based access



- network of cable, fiber attaches homes to ISP router
 - homes *share access network* to cable headend
- HFC: hybrid fiber-coaxial
 - Combines optical fiber and coaxial cable
 - asymmetric: up to 40 Mbps – 1.2 Gbps downlink rate, 30-100 Mbps uplink rate

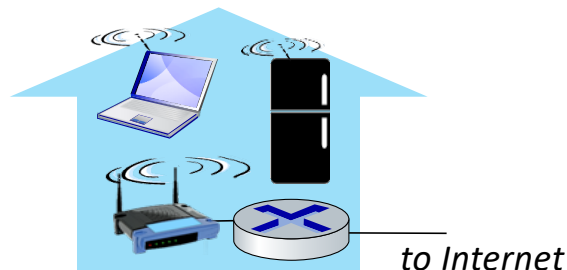
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

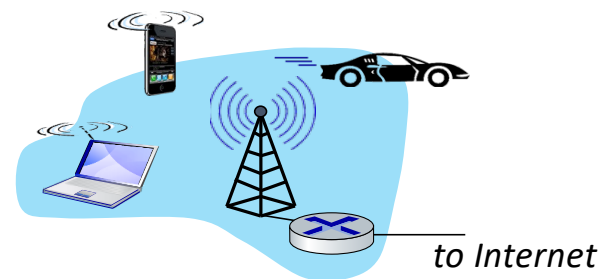
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n/ac (WiFi): 11, 54, 450 Mbps, 1Gbps transmission rate

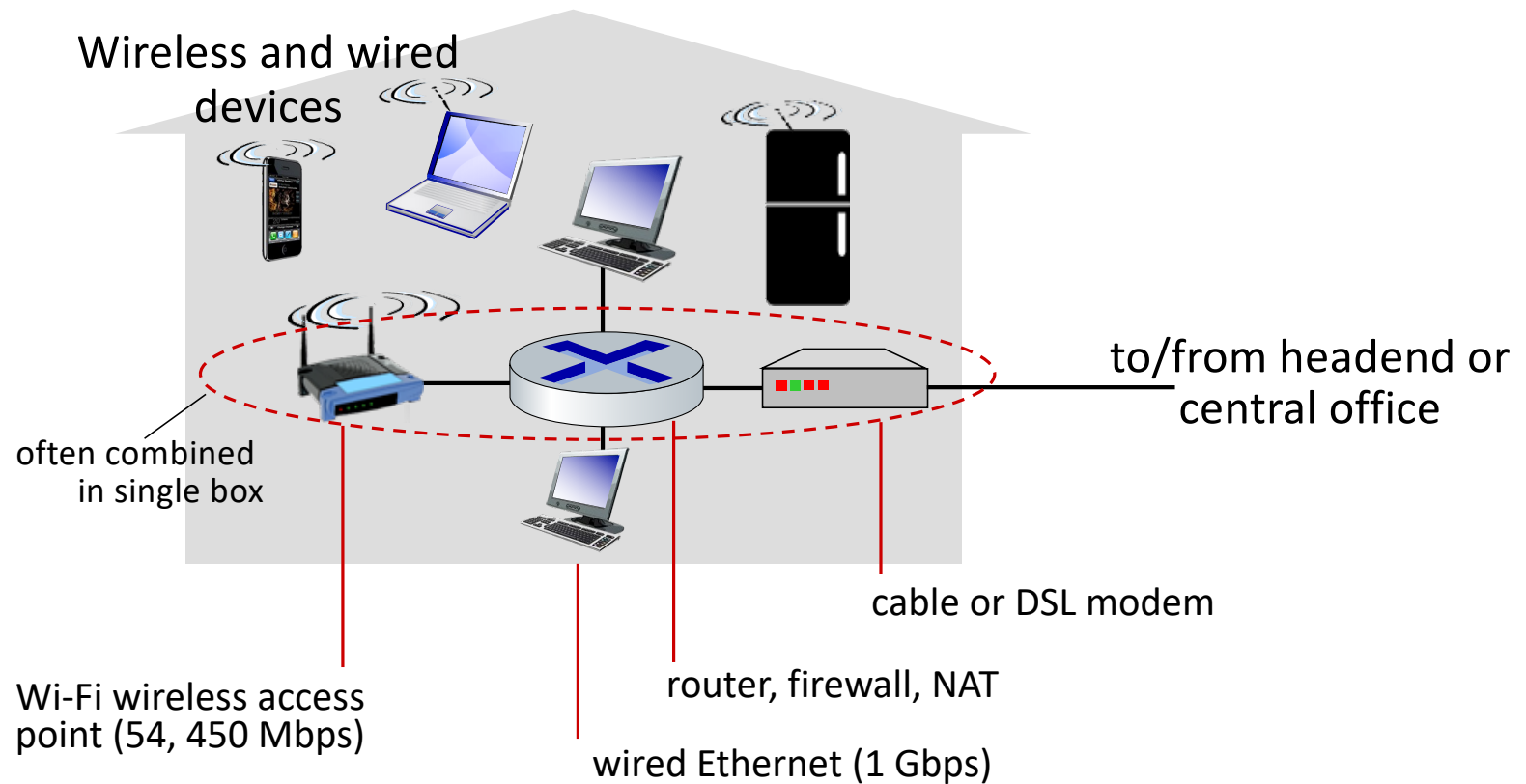


Wide-area cellular access networks

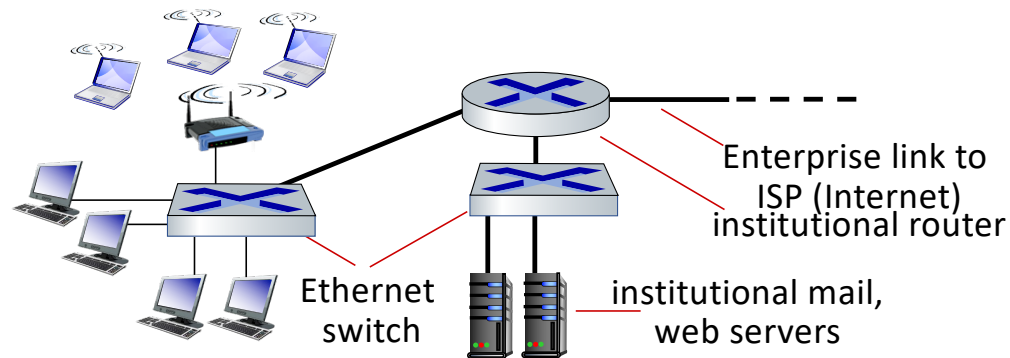
- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks



Residential/Home networks



Enterprise networks



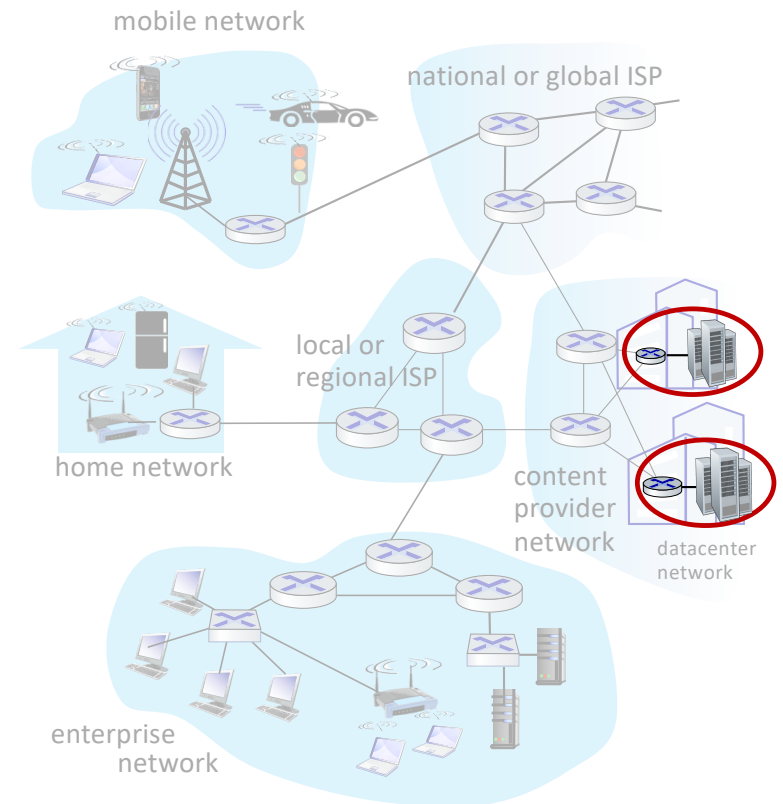
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - Wi-Fi: wireless access points at 11, 54, 450 Mbps

Data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)

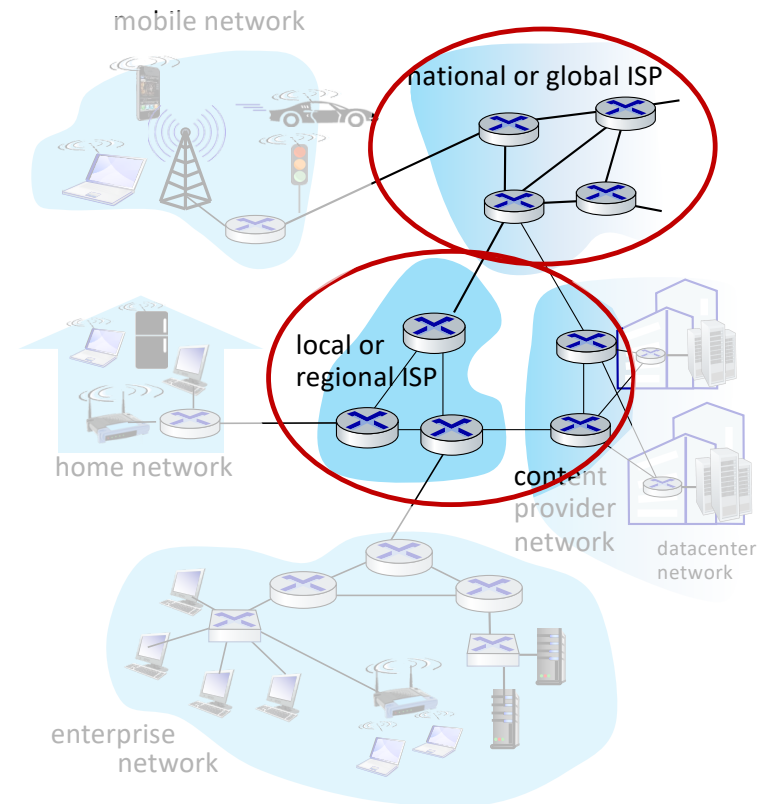


Overview/Roadmap

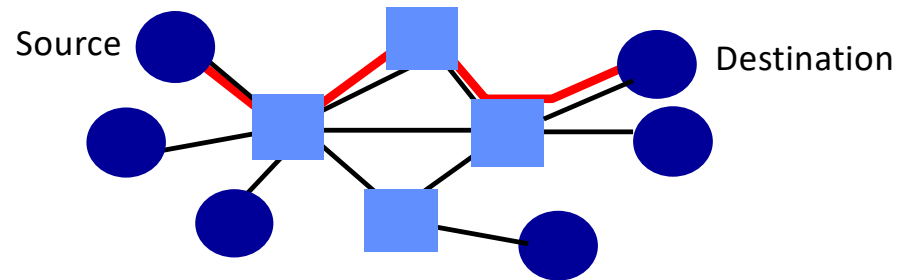
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The network core

- mesh of interconnected routers
- Q: How do routers forward packets from source to destination?

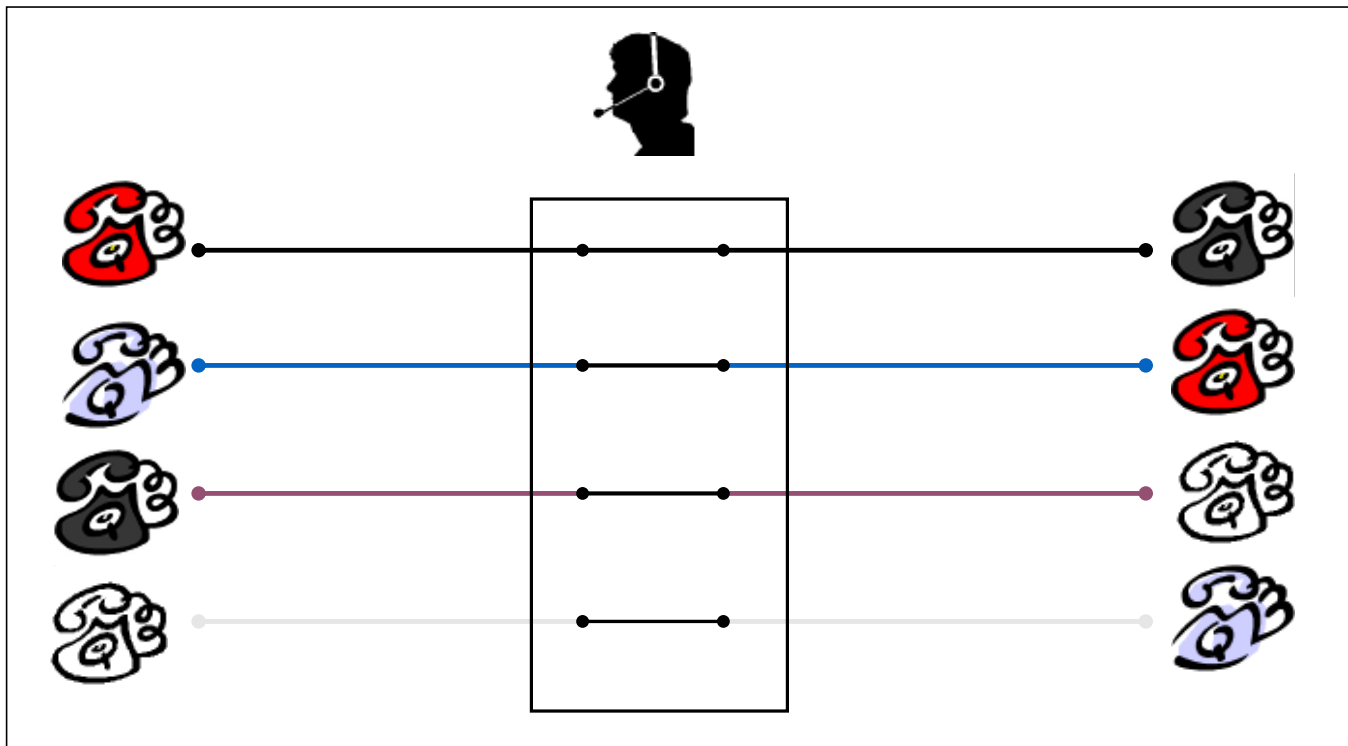


Circuit Switching



- Source establishes connection to destination
 - Nodes along the path store connection info
 - Nodes may reserve resources for the connection
- Source sends data over the connection
 - No destination address, since nodes know path
- Source tears down connection when done

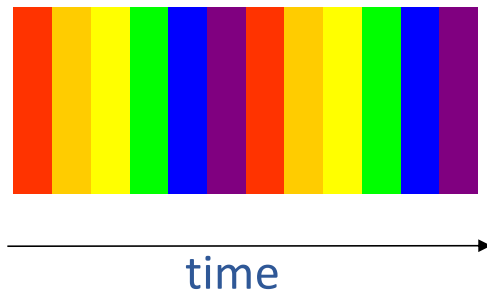
Phone Network Uses Circuit Switching



Circuit Switching: Multiplexing a Link

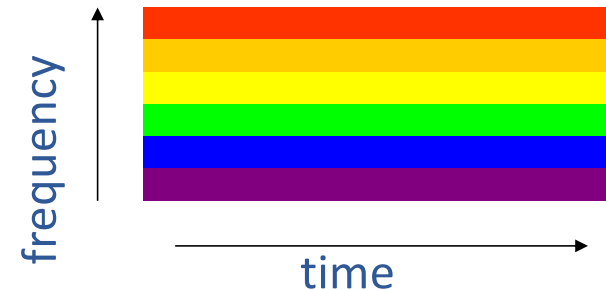
■ Time division

- Each circuit allocated with certain time slots



■ Frequency division

- Each circuit allocated with certain frequencies



Advantages of Circuit Switching

- **Guaranteed bandwidth**

- Predictable communication performance

- **Simple abstraction**

- Reliable communication channel between hosts
- No worries about lost or out-of-order packets

- **Simple forwarding**

- Forwarding based on time slot or frequency
- No need to inspect a packet header

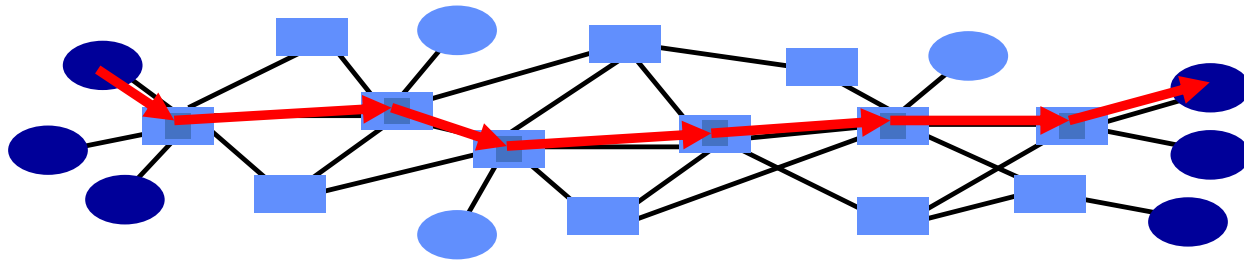
- **Low per-packet overhead**

- No IP (and TCP/UDP) header on each packet

Disadvantages of Circuit Switching

- **Wasted bandwidth**
 - Bursty traffic leads to idle connection during silent period
 - Unable to achieve gains from statistical multiplexing
- **Blocked connections**
 - Connection refused when resources are not sufficient
 - Unable to offer “okay” service to everybody
- **Connection set-up delay**
 - No communication until the connection is set up
 - Unable to avoid extra latency for small data transfers
- **Network state**
 - Network nodes must store per-connection information
 - Unable to avoid per-connection storage and state

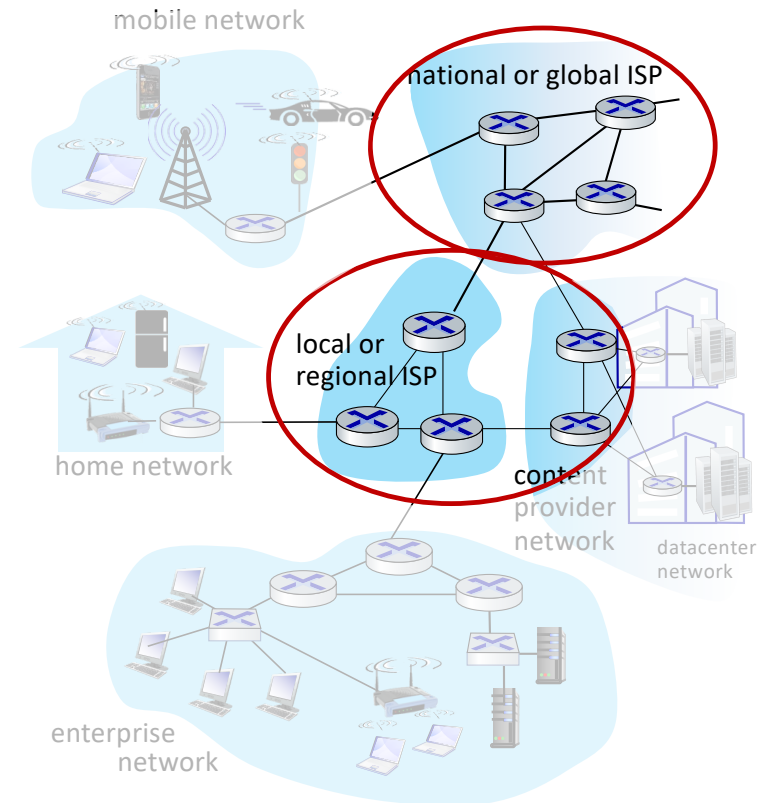
Packet Switching



- Traffic divided into packets
 - Each packet contains a header (with address)
- Packets travel separately in the network
 - Packet forwarding based on the header
 - Network nodes may store packets temporarily
- Destination reconstructs the message

The network core

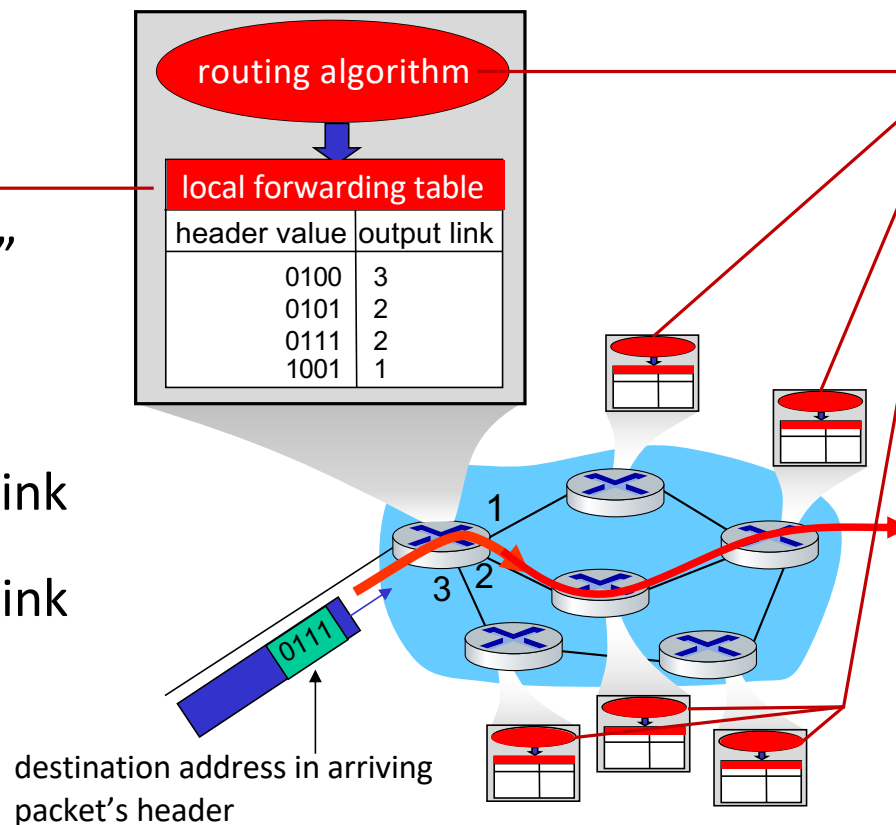
- mesh of interconnected routers
- **The Internet uses packet-switching:** hosts break application-layer messages into *packets*
 - network **forwards** packets from one router to the next, across links on path from **source to destination**
- **Best-effort** delivery
 - Packets may be lost
 - Maybe corrupted
 - May arrive out of order....



Two key network-core functions

Forwarding:

- aka “switching”
- *local* action: move arriving packets from router’s input link to appropriate router output link



Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms





Packet switching versus circuit switching

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior with packet-switching?**
 - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?